The human eye and brain present a fascinating problem for game developers. How do you make a game that can come close to matching the complexity of visual perception of the average human? With breathtaking graphics, of course. How do you make such games playable? Good question. Here's where the cards come in.

With the Radeon X800 and the GeForce 6800 Ultra, game developers have in their hands, the newest generation of programmable GPUs (Geometry Processing Unit). These have massive number crunching and processing abilities, cutting-edge memory technology, and performance optimisations.

At the cutting edge of the graphics card industry, nVidia has released its newest—the NV40-based GeForce 6800 Ultra. Current favourite ATi isn't standing idly by—they've just released the Radeon X800 XT (codenamed the R420), the successor to the hugely popular Radeon 9xxx family.

Computing power

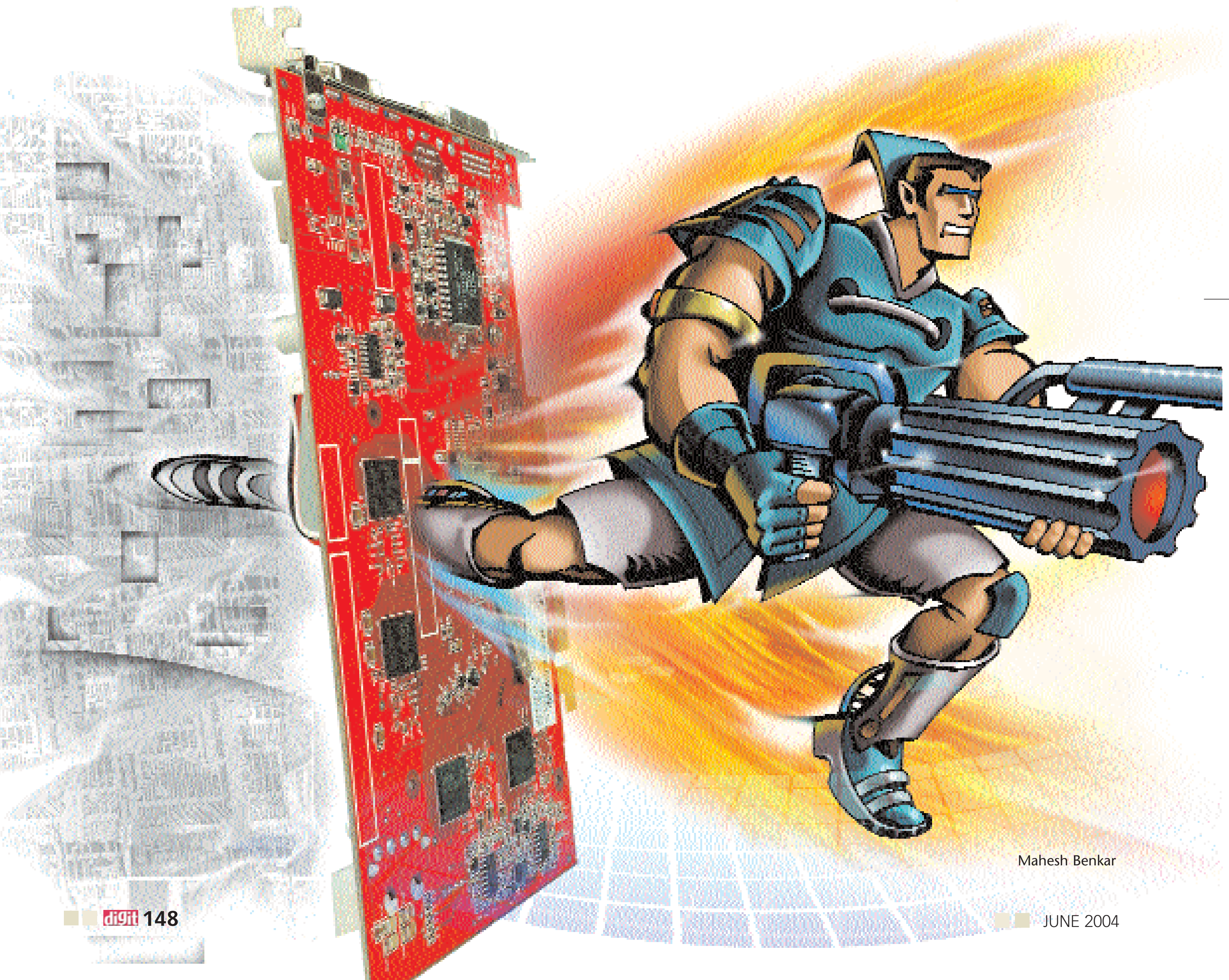
With the new GeForce 6 series family, nVidia has reworked much of the architecture to optimise the NV40's computational performance.

The GeForce 6800 also boasts of 220 million transistors, as against the 125 million on the GeForce FX 5950. This number is even more impressive when you look at the Pentium 4 Prescott's 125 million transistors. The ATi Radeon X800 XT comes packed with about 160 million transistors, as compared to the older Radeon 9800 XT's 110 million transistors. The Radeon X800 XT also scores on GPU clock frequency—it runs at a higher clock speed than all other comparable contenders. At a 512 MHz core clock, this ATi beast beats all the others.

Of course, core clock speeds are a bit hazy, and might well have changed by the time you get to buying the hardware. Part of this has to do with marketing—for instance, nVidia has announced plans to introduce a 6800 'GT' variant, a Porsche for the masses. Costing a little less than the 6800 Ultra, you can get the GeForce 6800 GT, with a slightly lower clock speed, but lots of graphics memory. Another special, possibly more expensive edition—the 'Extreme' GeForce 6800 Ultra—will run at 450 MHz GPU core clock frequency, with fast memory.

Drawing in parallel

More than the numbers, is the graphics pipelining detail. Well before the X800 XT and the NV40 came along, there's been considerable interest in the features and specs of the two families, and a lot of hype. Processing, and accurately render-



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